Zakaria Patel

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Skills

- Programming: Python, JavaScript, MatLab, Maple, C, Java
- Tools/Libraries: PyTorch, Keras, NumPy, Scikit-Learn, Pandas, AWS

Experience

Research Engineer **Ecomtent** *Toronto, ON, Canada* **05/2023 - 12/2023**

- Shipped the first AI content generation suite, which is currently used by 40+ e-commerce businesses
- Improved image generation quality by replacing existing components of the generation pipeline, increasing MRR from $300 to $6000
- Built the core prototype which allowed Ecomtent to be selected by Techstars as the top 1% of startups
- Increased software development productivity by 50% by refactoring codebase to align with software engineering principles
- Researched and developed a novel method for controllable image generation to provide improved customizability for users

Research Intern **Perimeter Institute** *Waterloo, ON, Canada* **09/2020 - 06/2022**

- Collaboratively designed a novel unsupervised machine learning method using Siamese neural networks to discover phase transitions in condensed matter systems
- Achieved accuracies over 95% in estimating phase boundaries of Ising systems, which competes well with existing supervised methods
- Reproduced the phase diagram of the well-studied Rydberg atom array in a fully unsupervised manner, with results highly consistent with prior research

Data Scientist **TRIUMF** *Vancouver, BC, Canada* **05/2021 - 08/2021**

- Applied machine learning-based particle-type classification with the goal of reducing backgrounds in neutrino oscillation measurements
- Improved ResNet-based convolutional neural network (CNN) for classification by analyzing training/testing curves and implementing new data transformations to improve generalization on the particularly challenging electron vs. gamma discrimination task by 5%
- Documented and presented experimental research results to a panel of over 15 researchers

Software Developer **AI Endurance** *Hamilton, ON, Canada* **06/2020 - 09/2020**

- Created production-level integration with Garmin API, allowing user files to be parsed into the appropriate backend data structure
- Developed and implemented a custom RNN and neural network training strategy to find an optimized training plan for a user after their custom machine learning model is trained. This RNN Training Plan optimization went into production.
- Improved convergence (finding an ideal plan for a user) from 60% to 95% and reduced cloud CPU usage by ~50% using the new custom RNN optimization method.

Education

MSc in Applied Computing **University of Toronto** *Toronto, ON, Canada* **09/2022 – 12/2023**

- Courses: Computational Imaging, Machine Learning, Natural Language Computing, Deep Learning
- cGPA: 4.0/4.0 (A+)

BEng in Engineering Physics **McMaster University** *Hamilton, ON, Canada* **09/2017 - 4/2022**

- cGPA: 11.7/12.0 (A+)

Research

Patel, Z., Merali, E, Wetzel, S. J. (2022). Unsupervised learning of Rydberg atom array phase diagram with Siamese neural networks. New Journal of Physics, 24 (11), 113021. https://doi.org/10.1088/1367-2630/ac9c7a

Patel, Z., & Rummel, M. (2021). Extremal Learning: Extremizing the output of a neural network in regression problems. https://doi.org/10.48550/ARXIV.2102.03626

Awards

- **Vector Scholarship in Artificial Intelligence** (Vector Institute, 2022)
- **P. Tan Academic Grant** (2021), **Leo Seto Scholarship** (2021), **Provost Honour Roll** (2020), President's Award (2017) (all McMaster)

Teaching

- **Introduction to Quantum Mechanics** (McMaster University)
- **Introduction to Computer Science** (University of Toronto)